Lab Experiment 8

Manipulating Spreadsheets



Image 1: logo of Microsoft Excel

Microsoft Excel is an electronics spreadsheet application that allows the user to do calculations, graphics, pivot tables, and basic programming. It can also be used for storing, organizing, and manipulating data.

Other interesting part of Microsoft Excel is that after data is entered in a cell of a spreadsheet, you can perform calculations on the numerical data and analyze the data to make informed decisions. When you make changes to one or more values of a cell, you can immediately see the effect of those changes in totals and charts that rely on those values.

The file document extension of Microsoft Excel is **.xlsx**

Microsoft Excel is made of spreadsheets organized together as a workbook.

Learning Outcomes

By the end of this lab, students will be able to:

1. Create and Format Cells and Spreadsheets

- Design and organize worksheets by entering, editing, and formatting data in cells.
- Apply formatting tools (e.g., fonts, borders, cell styles, conditional formatting) to enhance the visual structure of a spreadsheet.
- Utilize features like merging cells, adjusting row/column dimensions, and freezing panes for improved data presentation.

2. Use Math Operators and Statistical Functions

- Perform basic arithmetic operations (addition, subtraction, multiplication, division) using Excel formulas.
- Apply built-in statistical functions (e.g., AVERAGE, MAX, MIN, COUNT) to analyze data sets.
- Construct formulas using cell references to ensure dynamic calculation updates.

3. Create Charts and Pie Charts

- Generate and customize various types of charts, including bar, column, and pie charts, to visually represent data.
- Select appropriate chart types based on the data and intended message.
- Format chart elements such as titles, legends, and data labels to improve clarity and impact.

4. Apply Functions in Excel

- Identify and use essential Excel functions (e.g., IF, VLOOKUP, SUMIF) to solve problems and automate tasks.
- Combine functions to create nested formulas for more advanced data analysis.
- Troubleshoot and evaluate function-based formulas for accuracy and efficiency.

Lab Activity: Microsoft Excel Skills

In this lab experiment, students will develop essential skills in Microsoft Excel, focusing on foundational tools and techniques commonly used in data organization, analysis, and presentation. Through hands-on activities and guided practice, students will learn four main skills:

- **Skill 1: Creating and Formatting Cells and Spreadsheets**
  Students will gain experience entering and editing data, adjusting cell layouts, and applying formatting to enhance the readability and organization of spreadsheets.
- **Skill 2: Using Math Operators and Statistical Functions**
  This section introduces the use of basic arithmetic operators and common statistical functions such as SUM, AVERAGE, MIN, and MAX to perform calculations and summarize data.
- **Skill 3: Creating Charts and Pie Charts**
  Students will learn to visualize data effectively by creating and customizing various chart types, including column and pie charts, to communicate trends and comparisons clearly.
- **Skill 4: Applying Functions in Excel**
  Students will explore the use of Excel's built-in functions such as IF, VLOOKUP, and SUMIF to perform conditional calculations, automate data processing, and enhance spreadsheet functionality.

Skill 1 – Creating and Formatting cells and spreadsheet

Students will learn how to create and organize spreadsheets by entering and editing data, adjusting column widths and row heights, merging cells, and applying formatting such as fonts, borders, alignment, and colors. These skills will help enhance the clarity, readability, and overall presentation of their Excel worksheets, ensuring that data is well-structured and easy to interpret.

Procedure to complete skill 1:

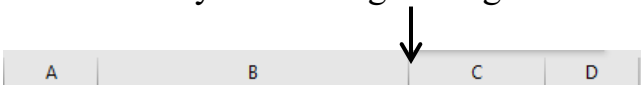
1. Open Microsoft Excel and file the Excel file as *Lab8StudentName.xlsx*
2. Click on cell A1 and type: TECH100 – Intro Engineering and Technology

   Click on cell A2 and type: GPA Calculator

   Complete the following cells with the following information:

   - B4: Items
   - B5: Class Assignments and Activities
   - B6: Project 1: ASL Project
   - B7: Project 2: Career Perspective
   - B8: Project Presentation
   - B9: Total

3. Go to the top of the cells and move the mouse to the border between cell B and C. When the mouse turns into a two ways arrow, double click on the border between cell B and C. This action makes the cells in column B to automatically fit the longest length of the cells within column B.

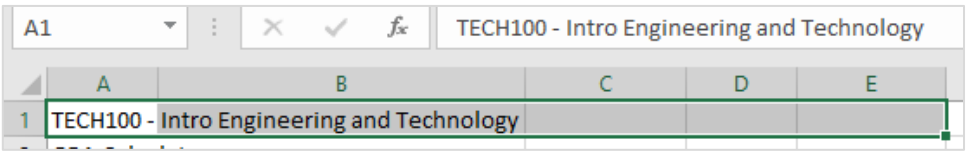


4. Complete the following cells with the following information:

   - C4: Grade Weight
   - C5: 50
   - C6: 15
   - C7: 25

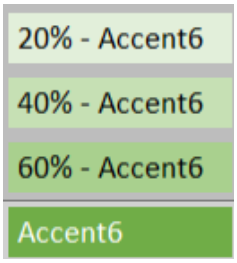
- C8: 10
- D4: Grade
- D5: 85
- D6: 76
- D7: 92
- D8: 88
- E4: Total Weight

5. Point to the middle of cell A1 to display the cross pointer. Click-hold on A1 and drag to the right up to E1 cell.



6. With the cells still selected, go to the **Home** tab, and in the **Alignment** group, click the **Merge & Center** tab.

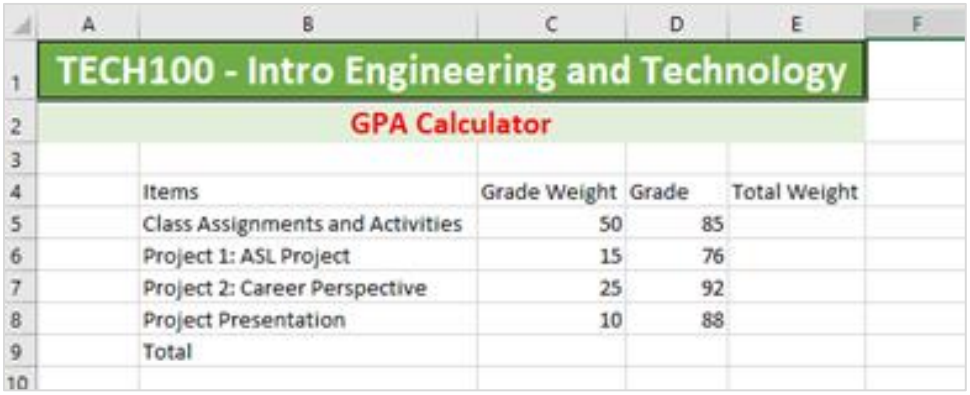
7. In the **Home** tab, in the **Styles** group, click the **Cell Styles** arrow, and click on the **Accent 6** from the list of styles.



8. With cell A1 through E1 still selected, change the font-size to **18**, the text style to **Bold**, and the font color to **White.**
9. Also click on the **Font Setting** arrow, which is the arrow located at the bottom-right of the **Font** group, to open the **Format Cells** window. In the Format Cells window, click on **Border** tab, select the double line Style and click on **Outline**

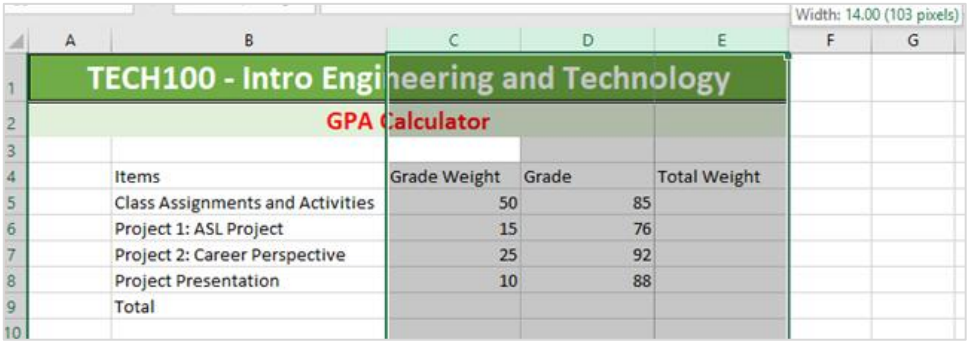
10. Click on cell A2 and hold down the left mouse and then drag to the right up to E2 cell.
11. In the **Home** tab, under the **Alignment** group, click the **Merge & Center**
12. In the **Home** tab, in the **Styles** group, click the More arrow, and click on the **20% - Accent6** from the list of styles.
13. With cell A2 through E2 still selected, change the font-size to **16**, the text style to **Bold**, and the font color to **Red.**

    The spreadsheet looks like the following image:



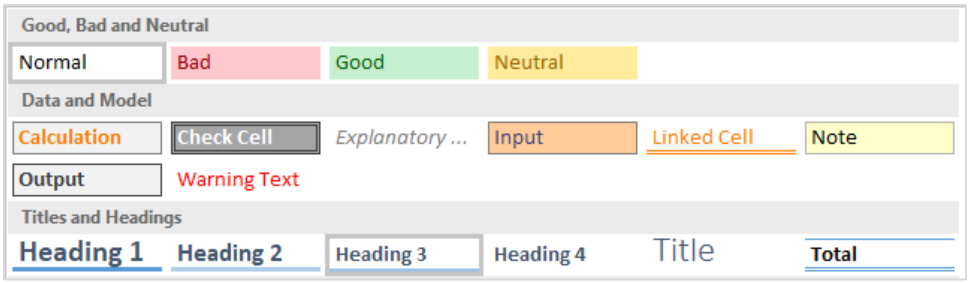
	A	B	C	D	E	F
1	TECH100 - Intro Engineering and Technology					
2	GPA Calculator					
3						
4		Items	Grade Weight	Grade	Total Weight	
5		Class Assignments and Activities	50	85		
6		Project 1: ASL Project	15	76		
7		Project 2: Career Perspective	25	92		
8		Project Presentation	10	88		
9		Total				
10						

14. Click on column C, hold and drag to the right up to column E. With column C, D, and E selected, go to the right-border of cell E, click and move the mouse to the right until it reads **Width: 14.00 (103 pixels)**

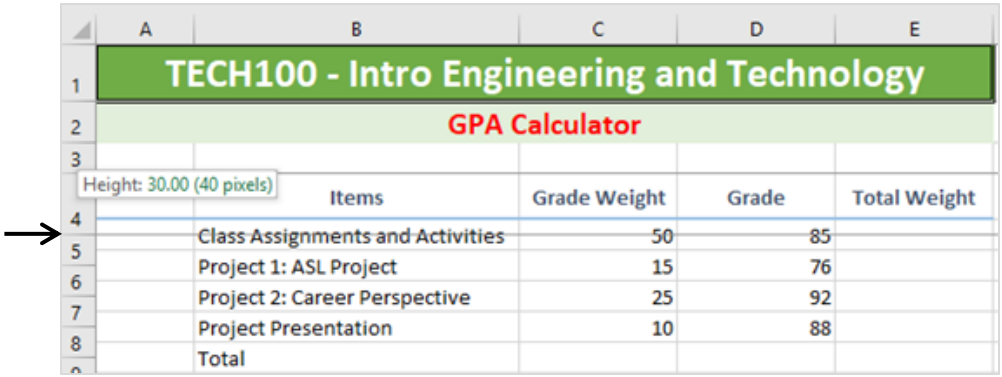


	A	B	C	D	E	F	G
1	TECH100 - Intro Engineering and Technology						
2	GPA Calculator						
3							
4		Items	Grade Weight	Grade	Total Weight		
5		Class Assignments and Activities	50	85			
6		Project 1: ASL Project	15	76			
7		Project 2: Career Perspective	25	92			
8		Project Presentation	10	88			
9		Total					
10							

15. Click on cell A4 and hold down the left mouse and then drag to the right up to E4 cell.
16. In the **Home** tab, in the **Styles** group, click the More arrow, and click on the **Heading 3** from the list of styles.



17. With cell A4 through cell E4 still selected, go to the **Alignment** group, and click on **Middle Align** and **Center** icons
18. Move the border of row 4 and 5, click and drag the mouse down until it reads **Height: 30.00 (40 pixels)**

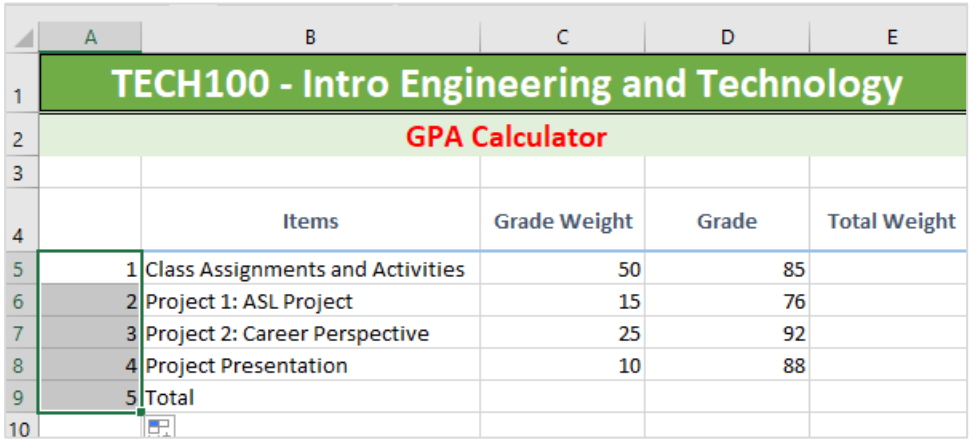


19. Save your work by clicking the Save icon at the top-left corner of Microsoft Excel, or by pressing Ctrl + S as a shortcut.

Autofill information in cells

Microsoft Excel allows users to autofill information and data in cells using the Hold-Drag technique.

20. Click on cell A5 and type **1**, and click on cell A6 and type **2.**
21. Click and hold on cell A5 and drag down to cell A6. With the two cells selected, move the mouse to the right-bottom of cell A5 until the mouse changes to a Black-Cross image. Once the mouse image is changed to a Black-Cross, click-hold and drag down to cell A9. This action automatically fills cells from A6 to A9 with an increasing count.



	A	B	C	D	E
1	TECH100 - Intro Engineering and Technology				
2	GPA Calculator				
3					
4		Items	Grade Weight	Grade	Total Weight
5	1	Class Assignments and Activities	50	85	
6	2	Project 1: ASL Project	15	76	
7	3	Project 2: Career Perspective	25	92	
8	4	Project Presentation	10	88	
9	5	Total			
10					

22. Move the mouse to the border between column A and B until the mouse pointer changes to a two-way arrow. When the mouse pointer changes to a two-way arrow, click-hold and drag to the left until it reads **Width: 4.00 (33 pixels)**
23. With column A still selected, click on the **Center** alignment to center the text in column A.

Skill 2 – Using math operators and statistics functions

Using mathematics operations

One of the powerful features of Microsoft Excel is the ability to perform mathematic operations, functions, and basic programming techniques. To perform these operations, we must go to function window and type the equal sign, = ,to let Microsoft Excel knows that an operation or function will be applied to the selected cell.

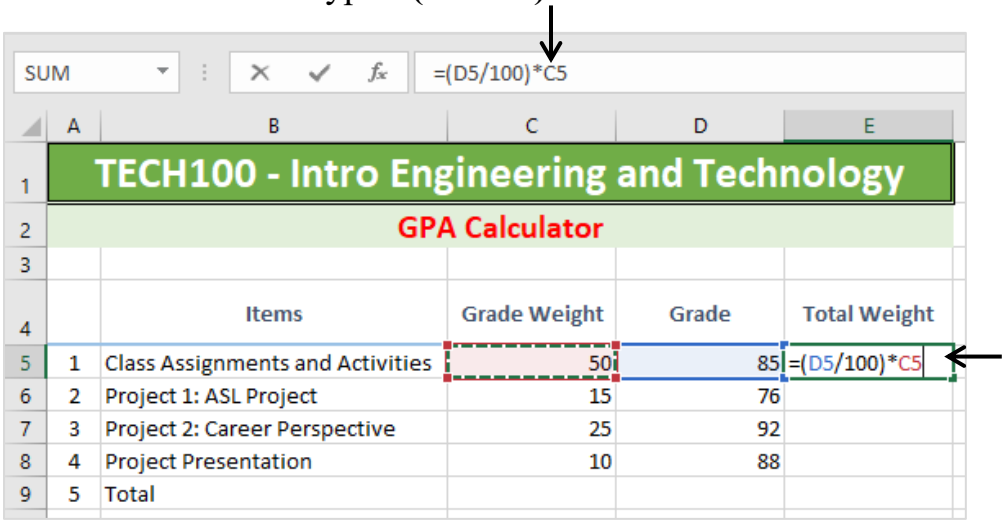


24. The formula to calculate the Total Weight of each Item is

$$Total\ Weight = \left(\frac{Grade}{100}\right) \times Grade\ Weight$$

To perform with calculation in Microsoft Excel, we use the cell location of Grade and Grade Weight

Click on cell E5 and type **=(D5/100)*C5**



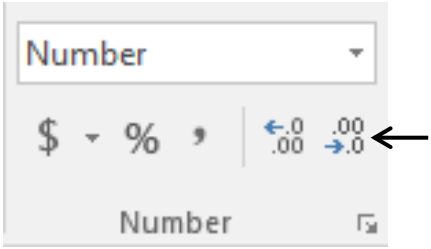
Once the formula is entered, press the **Enter** key.

25. Select cell E5 and move the mouse pointer to the right-bottom until the pointer changes to black-cross pointer.

26. Using the autofill technique, once the pointer changes to black-cross pointer, click-hold and drag down to cell E9. This technique will autofill the selected cells with the formula in cell E5.

	A	B	C	D	E	F
1	TECH100 - Intro Engineering and Technology					
2	GPA Calculator					
3						
4		Items	Grade Weight	Grade	Total Weight	
5	1	Class Assignments and Activities	50	85	42.5	
6	2	Project 1: ASL Project	15	76	11.4	
7	3	Project 2: Career Perspective	25	92	23	
8	4	Project Presentation	10	88	8.8	
9	5	Total				
10						
11						

27. With the E5 through E8 cells still selected, go to the **Number** group and increase the decimal to two decimal places. To increase decimal places, click on the **Increases Decimal** icon as many decimal positions as needed.



28. Save your work.

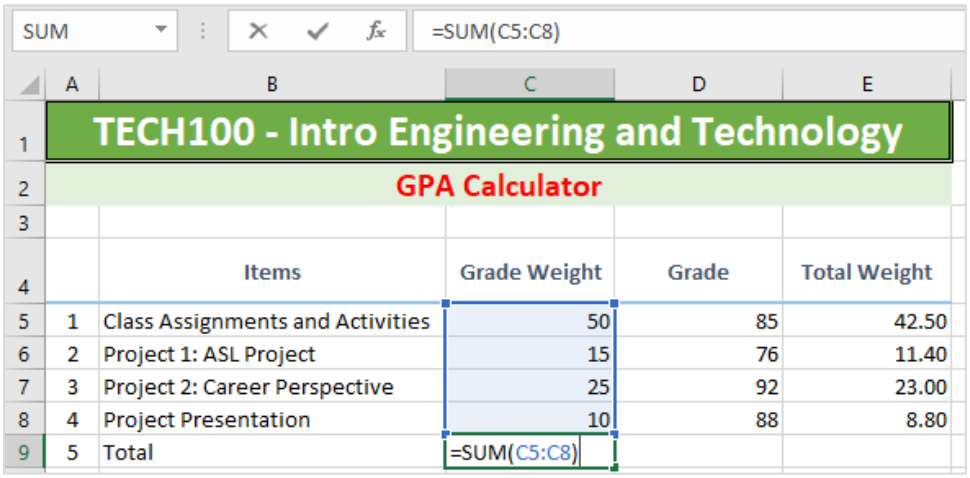
SUM and AVERAGE function

You can create your own functions, or you can use a ***function*** – a prewritten Excel formula that takes a value or values, performs an operation, and returns a value or values.

The ***SUM*** function is a built-in function in Excel that is categorized as a Math/Trigonometric function. It adds all numbers in a range of cells and returns the result.

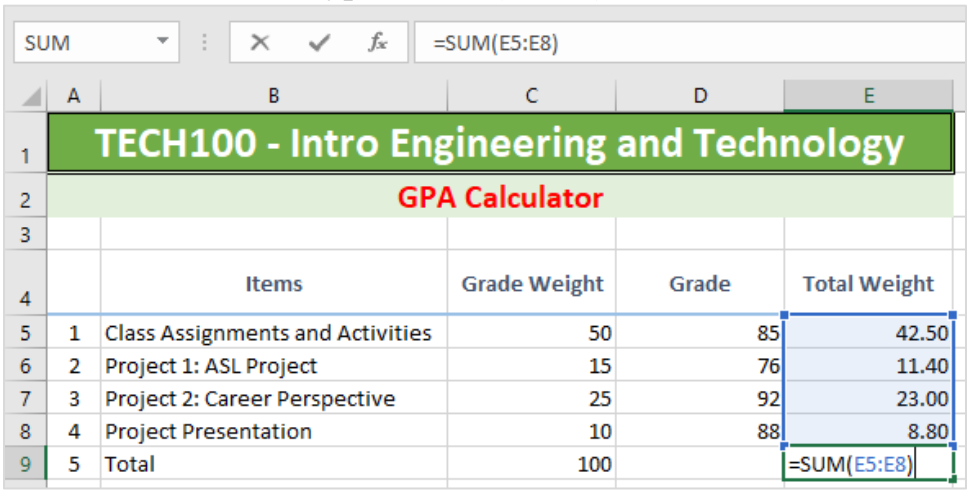
The ***AVERAGE*** function in Excel calculates the **average** (arithmetic mean) of a group of numbers set in a range. The **AVERAGE** function ignores logical values, empty cells and cells that contain text.

29. Click on cell C9 and type **=SUM(C5:C8)**



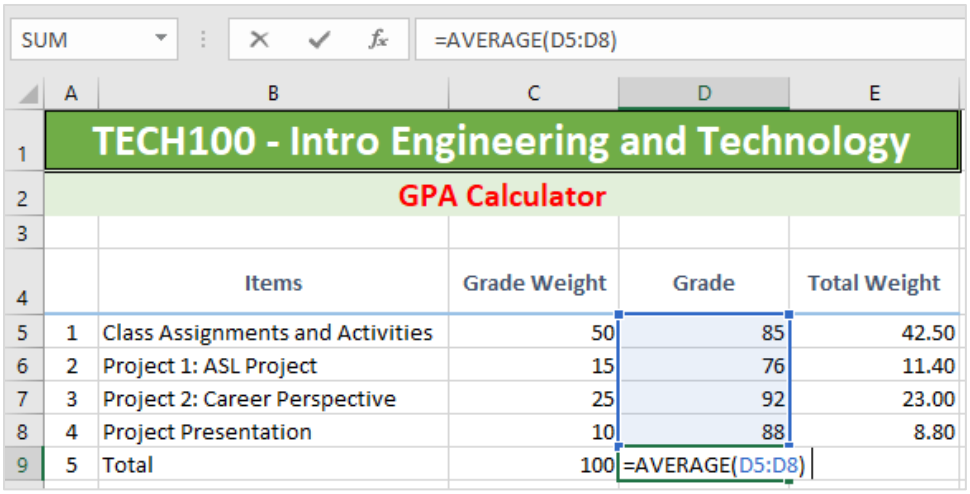
Once the equation is done, press **Enter**

30. Click on cell E9 and type =SUM(E5:E8)



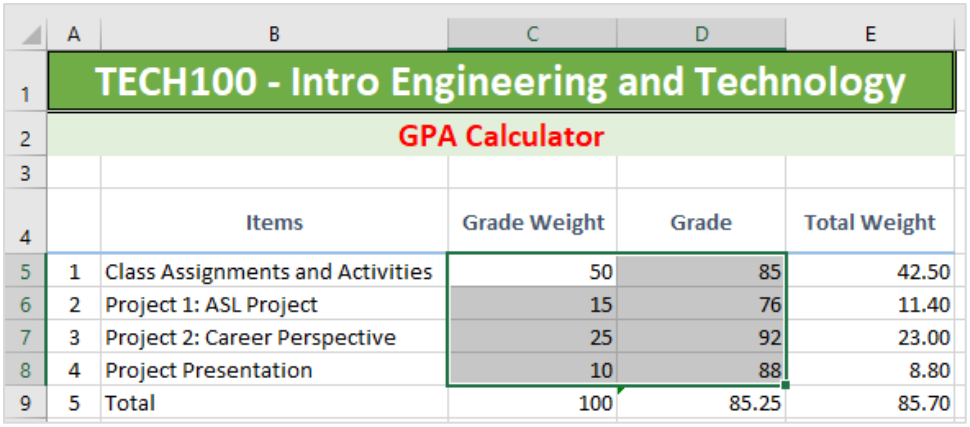
Once the equation is done, press **Enter**

31. Click on cell D9 and type =AVERAGE(D5:D8)



Once the equation is done, press **Enter**

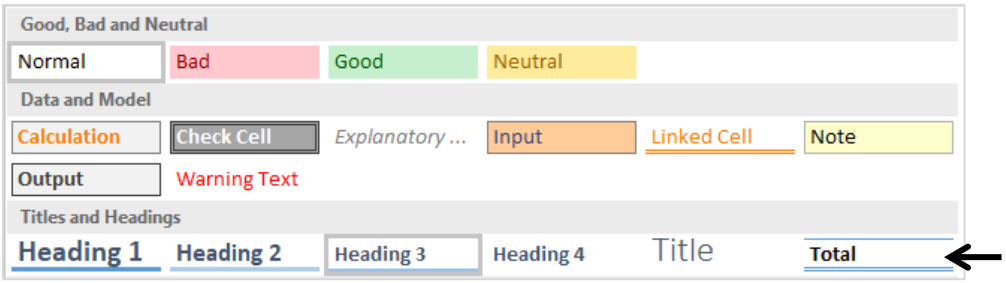
32. Click-hold on cell C5 and drag to the right up to cell D5 and drag down to D8. This selects the cells from C5 and D5 to C8 and D8



With the cells still selected, go to **Alignment** group and click on the **Center** icon.

33. Click-hold on cell E5 and drag down to cell E8. With the cells still selected, go to **Alignment** group and click on the **Center** icon, change the font color to **Red,** the text style to **Bold**¸ and fill the cells with background color of **lightgreen**.

34. Click-hold cell B9 and drag to the right up to cell E9. With the cells selected, go to the **Home** tab, in the **Styles** group, click the **More** arrow, and click on the **Total** from the list of styles.



35. With the cells still selected, go to **Alignment** group and click on the **Center** icon.
36. Click and cell B9 and change the alignment to the **left.**

The spreadsheet should look as:

	A	B	C	D	E
1	TECH100 - Intro Engineering and Technology				
2	GPA Calculator				
3					
4		Items	Grade Weight	Grade	Total Weight
5	1	Class Assignments and Activities	50	85	42.50
6	2	Project 1: ASL Project	15	76	11.40
7	3	Project 2: Career Perspective	25	92	23.00
8	4	Project Presentation	10	88	8.80
9	5	**Total**	**100**	**85.25**	**85.70**

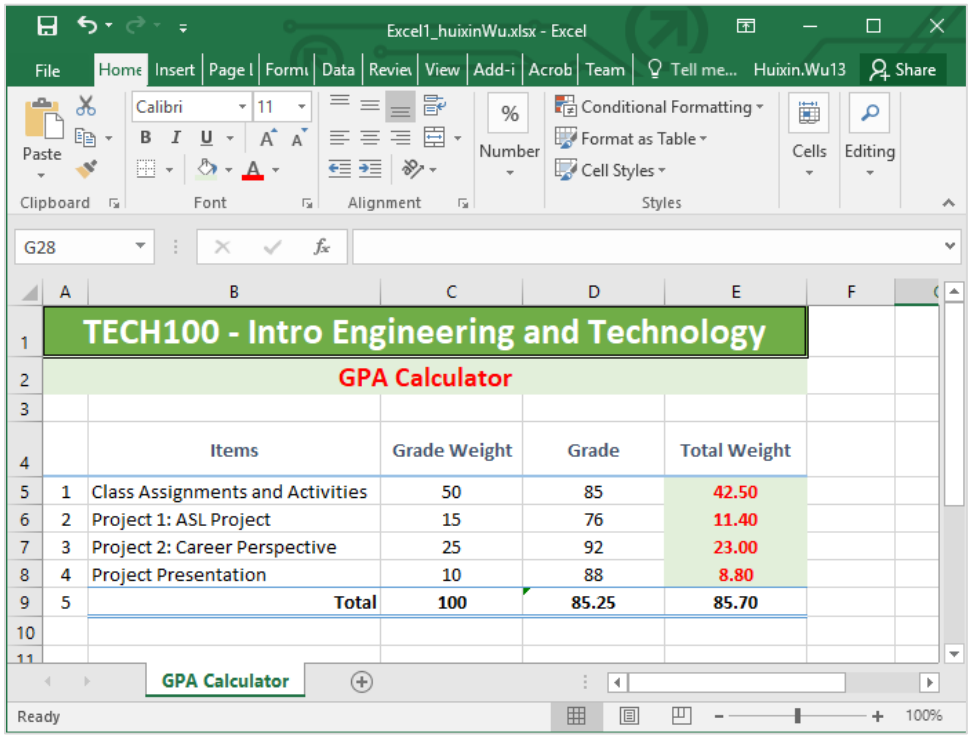
Changing the spreadsheet's name

37. Move the mouse pointer to the spreadsheet tab where says **Sheet1.** Right-click on **Sheet1** and click on **Rename** from the selection.
    Rename the sheet as **GPA Calculator**
38. Right-click GPA Calculator tab, select **Tab Color**, and from the color table, check the darkest green color, which is **Green, Accent 6, Darker 50%**

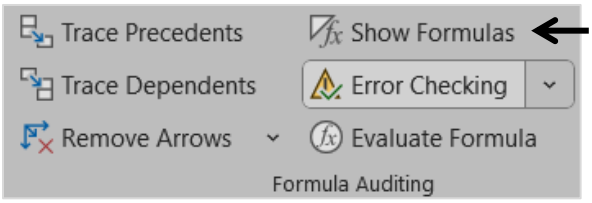


39. Save your work.

The final spreadsheet should look as:



To view the formulas, click on the **Formulas** tab, and click on **Show Formulas** from the **Formula Auditing** group.

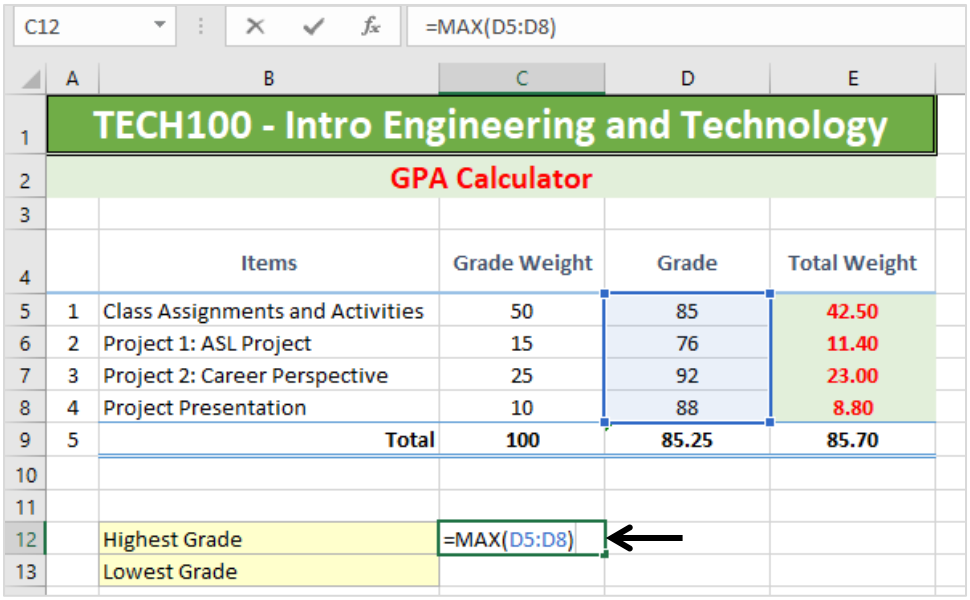


Minimum and Maximum formula

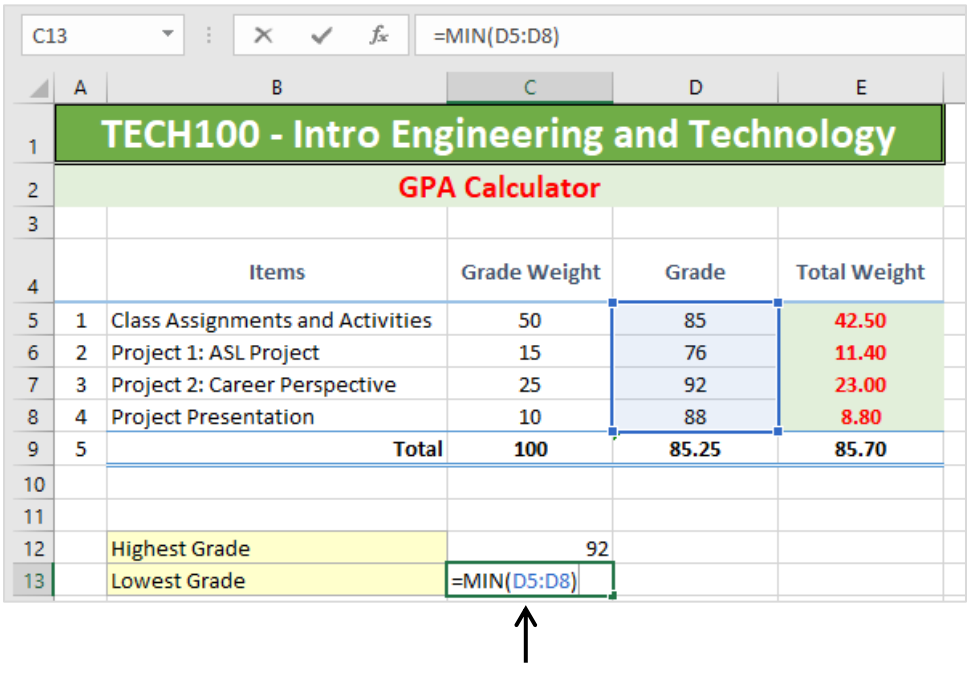
The **MIN** Excel function is used to find out the **minimum** value from a given set of data/ array. It returns the **minimum** value from a given set of numeric values. It will count numbers but ignore empty cells, text, the logical values TRUE and FALSE, and text values.

The Excel **MAX** Formula is used to find out the **maximum** value from a given set of data/ array. **MAX** function in Excel returns the highest value from a given set of numeric values**.** Excel **MAX** formula will count numbers but ignore empty cells, text, the logical values TRUE and FALSE, and text values.

40. Click on cell B12 and type *Highest Grade*
41. Click on cell B13 and type *Lowest Grade*
42. Select cell B12 and B13, go to **Home** tab, in the **Styles** group, click the **More** arrow, and select **Note** style. With still the cells selected, click on **Bold** from the **Font** list.
43. Click on C12 and type ***=MAX(D5:D8)*** This function returns the highest grade between D5 and D8.



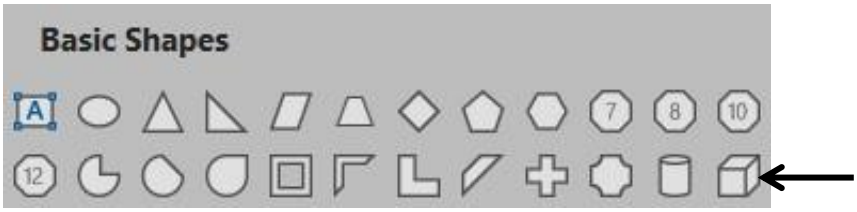
44. With cell C12 still selected, go to **Home** tab, in the **Styles** group, click the **More** arrow, and select **Good** style.
45. Click on C13 and type ***=MIN(D5:D8)*** This function returns the lowest grade between D5 and D8.



46. Select cell C13, go to **Home** tab, in the **Styles** group, click the **More** arrow, and select **Bad** style.
47. Save your work.

Working with Shapes

48. Go to **Insert** tab, click on **Shapes** icon, and select **Cube** icon from the **Basic Shapes** group.



49. With the mouse pointer active, drag the cube from cell B17 to C17 and down to C20.
50. With the cube still selected, go to **Shape Format** tab, click on **Shapes Styles** arrow, **Intense Effect – Green, Accent 6** from the **Theme Styles.**



51. Click on the front of the cube and type *Created by: type your student's name*
52. Select the cube, go to **Font** group, click on the font-family arrow and select **Brush Script Std** font family.
53. With the cube still selected, change the font size to **15** points, and set the font **Alignment** to **Middle Align** and **Center.**

The cube should look like the following image:



54. Save your work.

Skill 3 – Creating Charts and Pie Chart

In Microsoft Excel, you can create charts, including pie charts, by selecting your data and choosing a chart type from the Insert tab.

55. Create a new spreadsheet by clicking on the **New Sheet** icon



56. Double click on the new sheet and change the name to ***Assignment & Activities Grade*** and press **Enter** key.
57. Right-click on the new sheet, move the mouse pointer to **Tab Color**, and select the **Purple** color from the **Theme Color** window.
58. In the new sheet, ***Assignment & Activities Grade***, click on **A1** and type ***TECH100 – Intro Engineering and Technology.***
59. Click cell **A2** and type ***Class Assignments and Activities***
60. Click on the following cell and type:
    - B4: Assignment
    - B5: Creating Lab Report Using Microsoft Word
    - B6: Creating Dynamic Slide Shows using Microsoft PowerPoint
    - B7: Working and Formatting Spreadsheet in Microsoft Excel
    - B8: Creating Charts and Pie Charts in Microsoft Excel
    - B9: Digital Design using Multisim
    - B10: Creating Printed-Circuit-Board
    - B11: Introduction Engineering Programming using Python
    - B12: Control Flow and Making Decision in Python
    - B13: Creating 3D figures using autoCAD
    - B14: 3D figure printing
    - B15: AVERAGE ASSIGNMENTS & ACTIVITIES GRADE

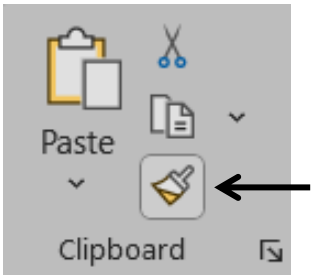
61. Move the mouse pointer between column **B** and **C**, double click between column **B** and **C** to automatically adjust the width of column B.
62. Click cell **A4** and type ***Activity***
63. Click cell **A5** and type 1
64. Click cell **A6** and type 2
65. Select cell A5 and A6, move the mouse pointer to the bottom-right of cell A6. Once the mouse pointer changes to **Plus** sign, click-hold and drag down to cell **A14**. This automatically updates the cells count from 1 to 10.

66. With cells **A5** through **A14** still selected, go to **Font** group and click on **Middle Align** and **Center** Alignment.
67. Click on cell **C4** and type ***Grade***
68. Click on the following cell and type:
    - C5: 50
    - C6: 60
    - C7: 70
    - C8: 90
    - C9: 70
    - C10: 80
    - C11: 70
    - C12: 90
    - C13: 100
    - C14: 100
69. Use the click-hold techniques and select from **C5** to **C14.** With cells **C5** through **C14** still selected, go to the **Font** group and click on **Middle Align** and **Center** Alignment.

Format Painter

Format Painter. The **Format Painter** is one of the most underused features of **Excel**. The **Format Painter** copies **formatting** from one place and applies it to another

70. Click on the **GPA Calculator** tab to go to the GPA Calculator sheet.
71. Select on the title, which from it is the merged cells from A1 to E1, and the subtitle, which from it is the merged cells from A2 to E2. With the title and subtitle selected, go the **Home** tab, click on the **Format Painter** icon from the **Clipboard** group.



72. With the title and subtitle cells active, click on the **Assignment & Activities Grade** tab's sheet. In the Assignment & Activities Grade's sheet, click-hold on cell **A1**, drag through **C1**, and down to cell **C2.** This method applies the same formatting in the title and subtitle of GPA Calculator sheet to cells **A1** to **C1** and cells **A2** and **C2** in Assignment & Activities Grade sheet.
73. Click on **GPA Calculator** tab again. Use the click-hold technique and select cells from **A4** to **E4.** Go to the **Clipboard** group and click on the **Format Painter** icon.
74. With cells **A4** and **C4** selected, click on the **Assignment & Activities Grade** tab's sheet. In the Assignment & Activities Grade's sheet, click-hold on cell **A4** and drag through **C4**.
75. In the **Assignment & Activities Grade** sheet, click on cell **C15** and type **=AVERAGE(C5:C14)**.
76. Click on **GPA Calculator** tab again. Use the click-hold technique and select cells from **B9** to **E9.** Go to the **Clipboard** group and click on the **Format Painter** icon.
77. With **B9** and **E9** selected, click on the **Assignment & Activities Grade** tab's sheet. In the Assignment & Activities Grade's sheet, click-hold on cell **B15** and drag through **C15**.
78. Save your work.

Creating Charts

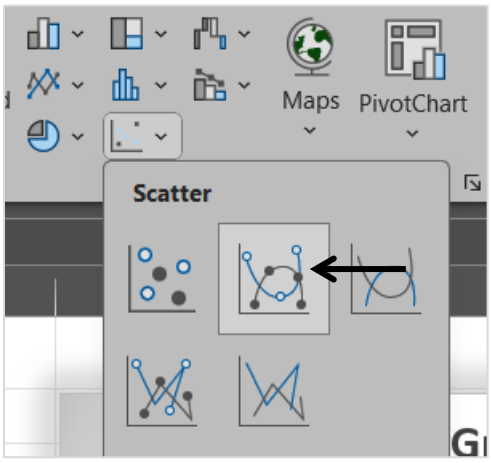
To create charts in Excel, we need to select the values for the *x-axis* and the *y-axis*, and then select the type of charts that we want to display the selected data.

To create a chart from blank data:

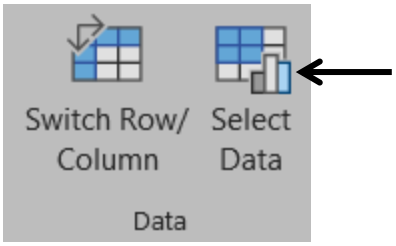
79. Go to **Insert** tab, under the **Charts** group, click on the **Insert Scatter (X,Y) or bubble Charts**



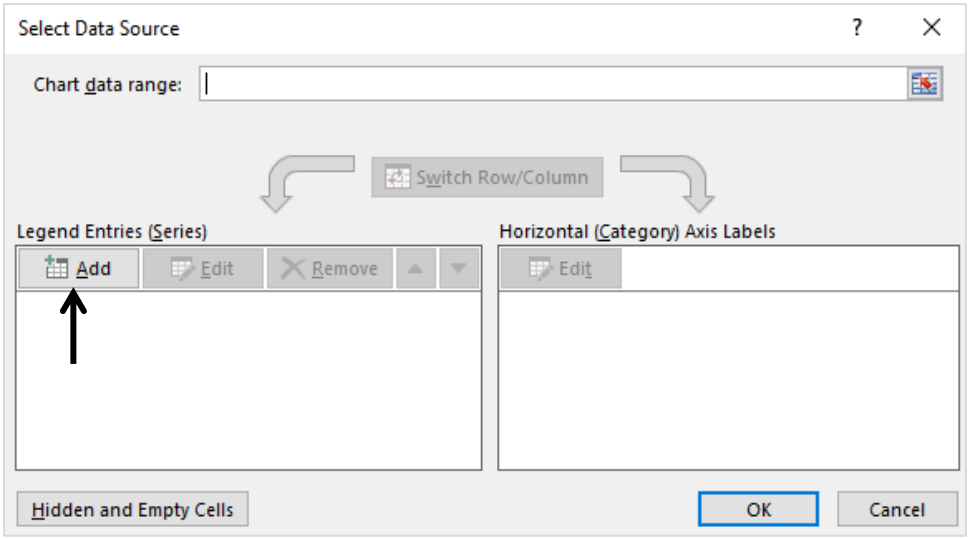
80. From the **Scatter** group, click on **Scatter with Smooth Lines and Markers**.



81. With the chart still selected, go to **Chart Design** tab and click on **Select Data** icon from the **Data** group.

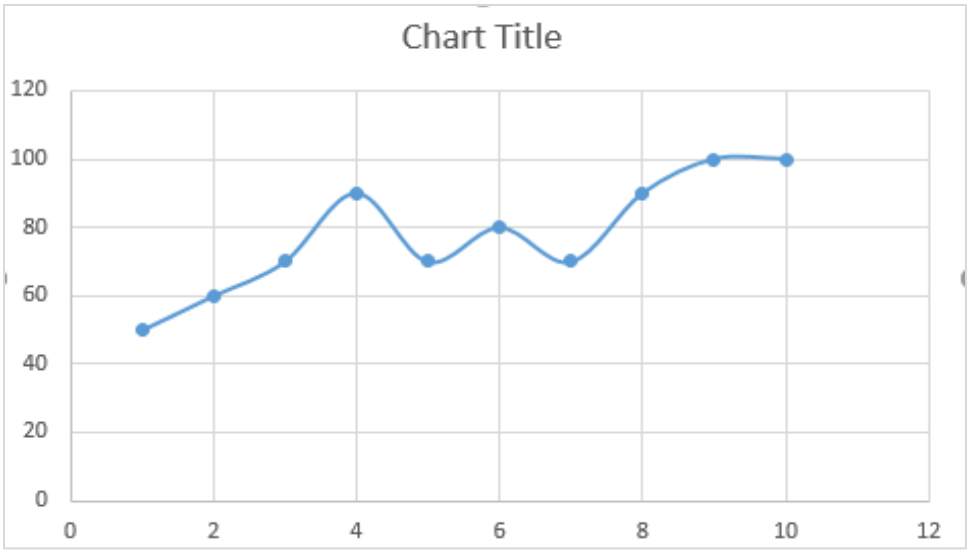


82. In the **Select Data Source** dialog box, click on **Add** icon under the **Legend Entries (Series)**



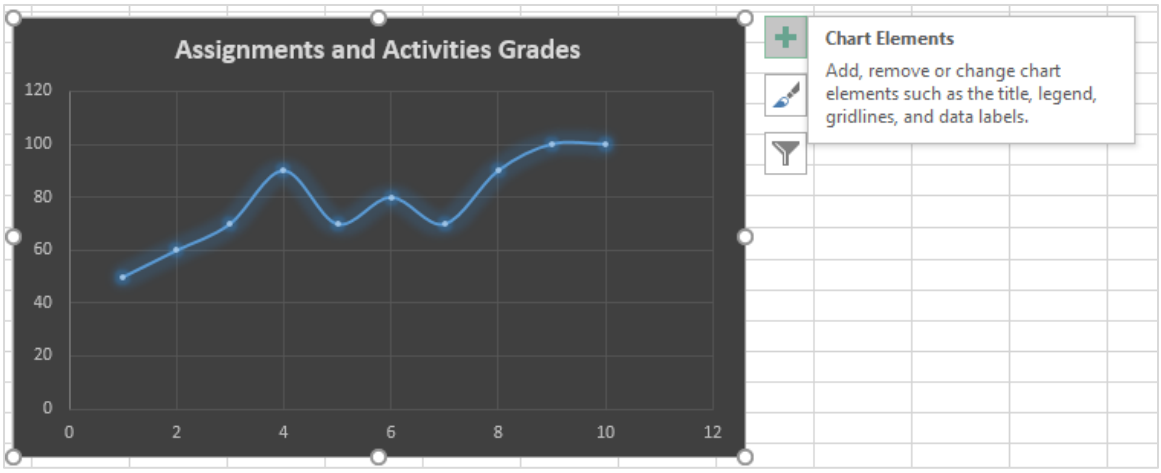
83. In the **Edit Series** window, click on the **Series X Values** cell, and select the X values from the sheet. If you want the *x-values* to be the ***Activity*** number, then we select from cell **A5** to cell **A14.**
84. In the **Edit Series** dialog box, click on the **Series Y Values** cell, and select the Y values from the sheet. If you want the y-values to be the ***Grade*** number, then we select from cell **C5** to cell **C14.**
85. When the x and y values are selected, click the **OK** tab to close the *Edit Series* window, and then click the **OK** tab from the *Select Data Source* window to close the current window.

    The chart should look like the following image:

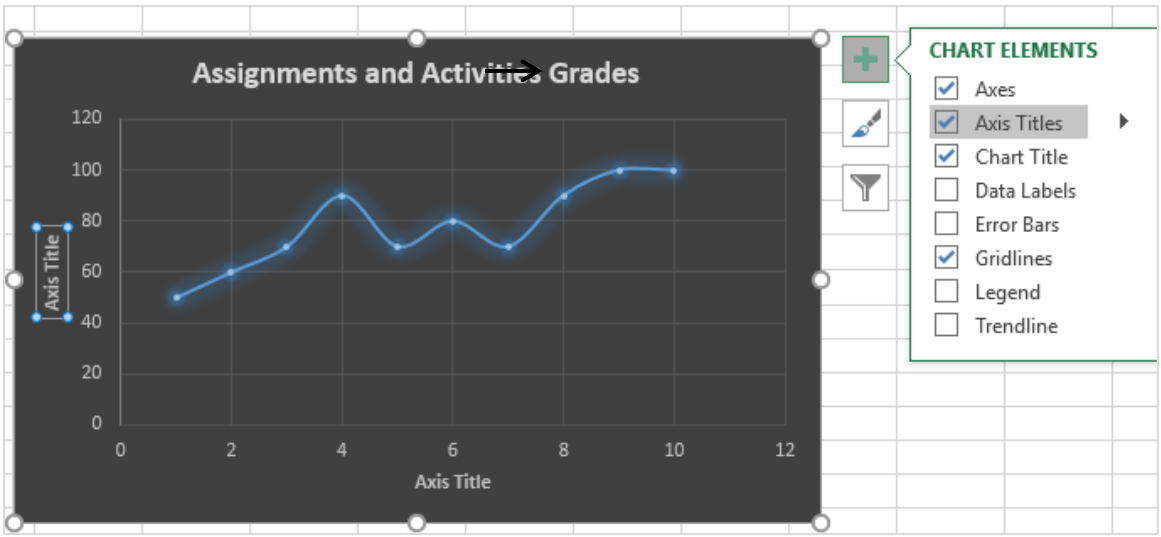


Formatting a chart

86. With the chart still selected, go to **Design** tab, click on **Chart Styles** arrow, and select **Style 6** style.
87. To change the chart title, double-click the chart title and type ***Assignments and Activities Grades***
88. To add x and y axis label, click on the chart, and when the right elements appear on hover, click on the **Chart Elements** icon.

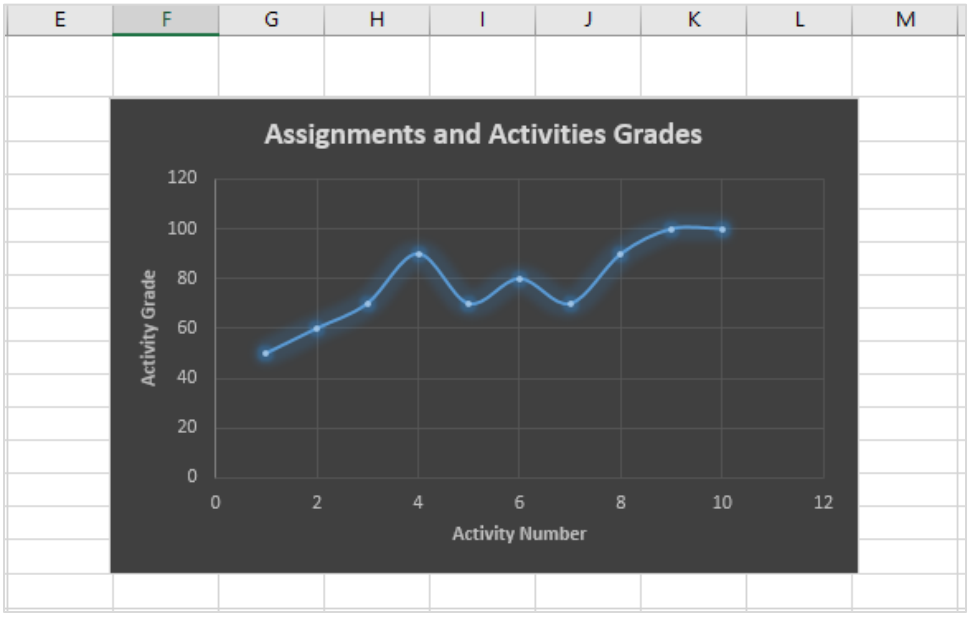


89. From the **Chart Elements** list, check the **Axis Titles**



90. To change the *x axis* label, double-click on *x Axis Title* and type ***Activity Number***
91. To change the y *axis* label, double-click on *y Axis Title* and type ***Activity Grade***
92. Click-hold the chart and move to the top of the sheet where the top-right corner aligns with cell F2.

The chart should look like the following image:

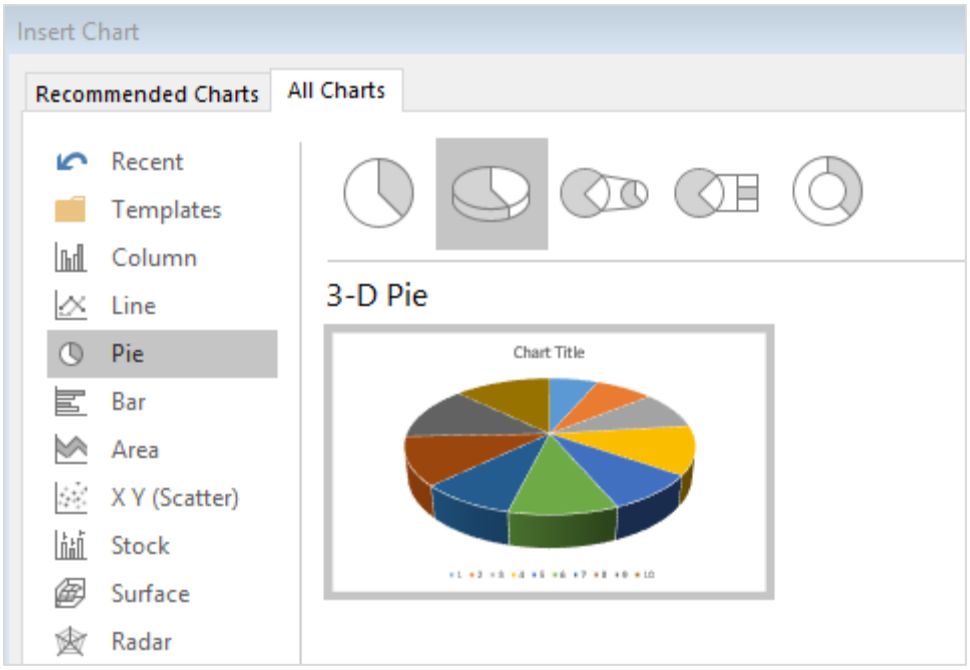


93. Save your work.

Creating Pie Chart

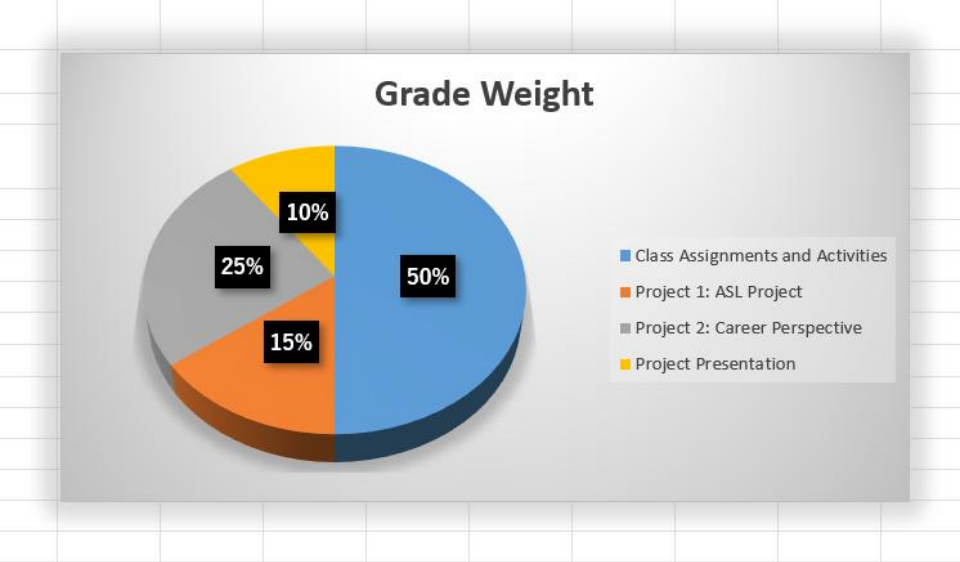
A pie chart displays the relationship of parts to a whole. For example, we can use the pie chart to display how offend we score different grades.

94. From the **GPA Calculator** sheet, select the range of cells from **C5** to **C8**, hold down the **Ctrl** key from the keyboard, and select the range of cells from **B5** to **B8**.
95. Go to the **Insert** tab, under the **Charts** group, click the **Recommended Charts** button.
96. From the **Insert** table, click on **All Charts** tab, click on the **3-D Pie**, and then click **OK.**



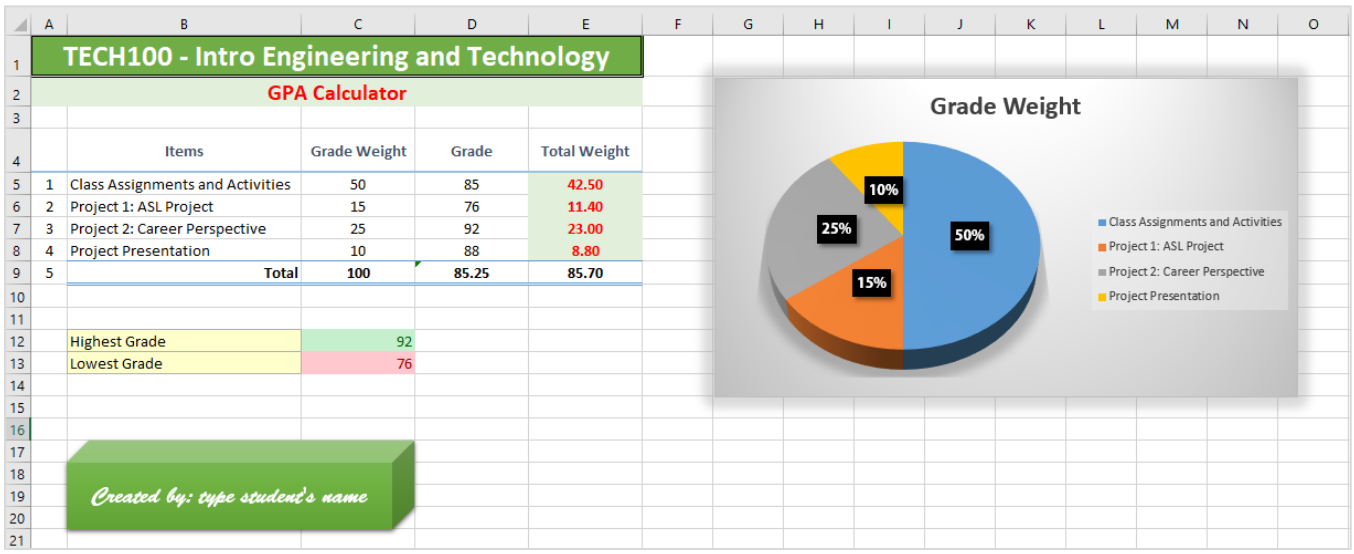
97. With the pie chart still selected, click on **Chart Styles** arrow, and select **Style 3** style.
98. Double click on the pie chart title and type *Grade Weight*
99. Click on one of the percentage numbers in the pie chart to select all the percentage data.
100. With all the percentage data selected, click the **Format** tab, click on the **Shape Fill** icon, and click on the **Black** color.
101. With all the percentage data still selected, go to the **Home** tab, change the font family to **Adobe Gothic Std** from the **Font** group.
102. Click on the **Legend** window, click the **Format** tab, go the **Size** group and change the **Width** to **5.5"**
103. Click on the corner of the pie chart to select the whole pie chart. With the pie chart selected, click the **Format** tab, click on **Shape Effects,** move the mouse pointer to **Glow** icon, and click on **Gray 50%, 18pt glow, Accent Color 3.**

The chart should look like the following image:



104. With the pie chart still selected, click-hold the pie chart and move it to the top of the sheet, where the upper-left corner aligns in cell **G2.**

GPA Calculator sheet should look as:



TECH100 - Intro Engineering and Technology

GPA Calculator

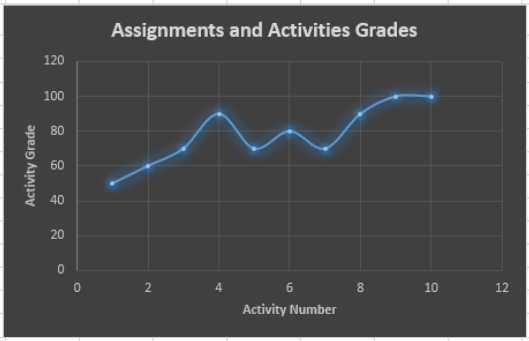
	Items	Grade Weight	Grade	Total Weight
1	Class Assignments and Activities	50	85	42.50
2	Project 1: ASL Project	15	76	11.40
3	Project 2: Career Perspective	25	92	23.00
4	Project Presentation	10	88	8.80
5	Total	100	85.25	85.70

Highest Grade	92
Lowest Grade	76

Created by: type student's name

Assignment & Activities Grade sheet should look as:

	A	B	C
1	TECH100 - Intro Engineering and Technology		
2	Class Assignments and Activities		
3			
4	Activity	Assignment	Grade
5	1	Creating Lab Report Using Microsoft Word	50
6	2	Creating Dynamic Slide Shows using Microsoft PowerPoint	60
7	3	Working and Formatting Spreadsheet in Microsoft Excel	70
8	4	Creating Charts and Pie Charts in Microsoft Excel	90
9	5	Digital Design using Multisim	70
10	6	Creating Printable-Circuit-Board	80
11	7	Introduction Engineering Programming using Python	70
12	8	Control Flow and Making Decision in Python	90
13	9	Creating 3D figures using autoCAD	100
14	10	3D figure printing	100
15		AVERAGE ASSIGNMENTS & ACTIVITIES GRADE	78
16			



105. Save your work.

Skill 4 – Applying Functions is Excel

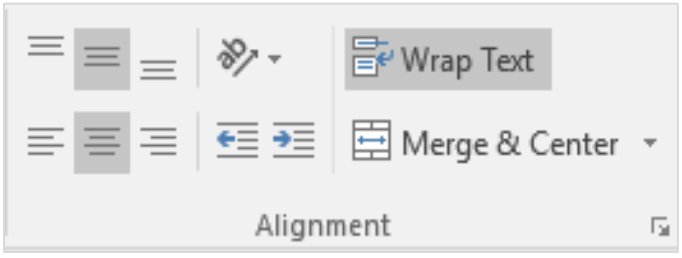
A **function** is a **predefined formula** that performs calculations using specific values in a particular order. Excel includes many common functions that can be used to quickly find the **sum**, **average**, **count**, **maximum value**, and **minimum value** for a range of cells. In order to use functions correctly, you'll need to understand the different **parts of a function** and how to create **arguments** to calculate values and cell references.

The **TODAY( )** function returns the serial number of the current date.

The **NOW( )** function returns the serial number of the current date and time.

Displaying current date and time

106. Go to **GPA Calculator** sheet, click on cell **B15** and type *Today Date and Time*
107. Click on cell **C15** and type **=NOW( )**
108. Select cell **B15** and **C15,** in **Home** tab, go to the **Font** group, change the font family to **Adobe Gothic Std** and the font-size to **10** pts, and click on the **Fill Color** icon and select the color **White, Background 1, Darker 15%**
109. Go to **Assignment & Activities Grade** sheet, click on cell **A18** and type *Today's Date.* If the text does not fill the whole cell, select cell **A18**¸ go to **Alignment** group and click on **Wrap Text**, **Middle Align,** and **Center** icons.



110. Click on the cell **B18** and type **=TODAY( ).** With cell **B18** still selected, go to **Alignment** group, click on **Middle Align** and **Align Left** icons.

111. Select cells **A18** and **B18**. With the cells selected, click on the **Styles** arrow, and click on the **Check Cell** icon.

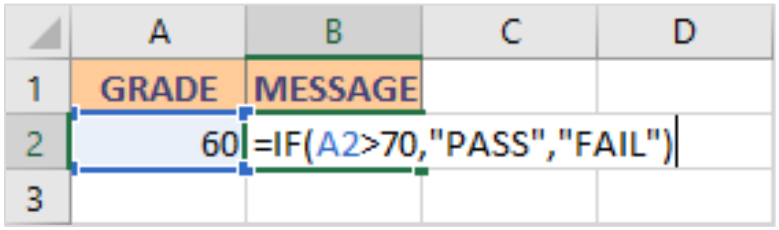
Applying IF function

The IF( ) function is a logical function that checks whether criteria are met and then returns one value when the condition is TRUE and another value when the condition is FALSE.

The syntax, or sentence structure, which is the way the commands are organized in the formula, of an Excel IF statement is:

=IF(logic_test, value_if true, value_if_false)

The **logic_test** uses condition to check if the statement is true of false. For example, we want to use cell B2 to check if cell A2 is greater than 70, then the of the **logic_test** of the IF statement is **A2>70.** If the **logic_test** is TRUE, then cell B2 will display "PASS", otherwise, cell B2 will display "FAIL". The complete logical statement will be:



In this activity we are going to check how many grades we have scored lower than 80.

112. Click on cell **D4** and type **Grade Less than 80.**
113. Select cell **D4**, go to **Alignment** group and click on **Wrap Text**, **Middle Align,** and **Center** icons. Also go to in between column **D** and **E,** click-hold in between column D and **E** and change the width to **Width:12.00 (89 pixels)**
114. Click on cell **D5** and type **=IF(C5<80,"LOW","GOOD")**, and press **Enter** key
115. The result for cell **D5** should say **LOW**

We are going to apply the same formula in cell **D5** to cells from **D6** to **D14**. One way to do so is copy cell **D5** and paste the formula from **D6** to **D14**. But Excel can help us to copy and paste data or information in another cells by using the click-hold and drag technique.

116. Click on cell **D5** and move the mouse pointer to the bottom-right corner of the cell until the pointer changes to a **Plus** sign. Click-hold and drag down to **D14.**

The cells should look like the following image:

Grade Less than 80
LOW
LOW
LOW
GOOD
LOW
GOOD
LOW
GOOD
GOOD
GOOD

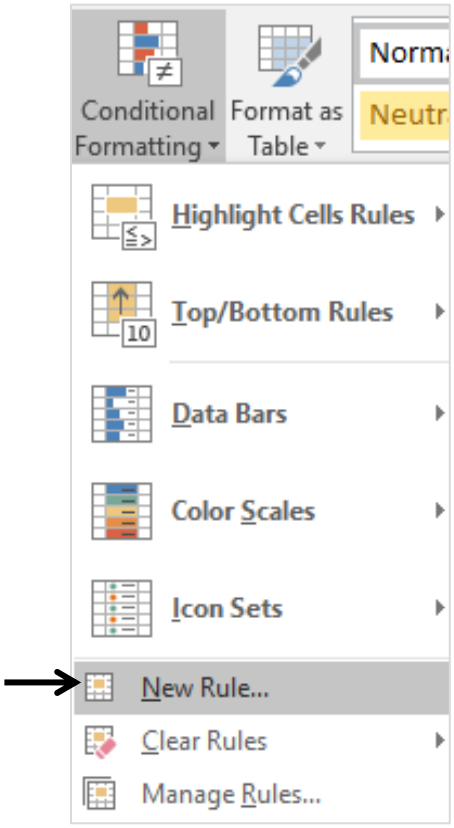
Applying Conditional Formatting

Conditional Formatting is a tool that allows you to apply formats to a cell or range of cells and have that **formatting** change depending on the value of the cell or the value of a formula. For example, you can have a cell appear bold only when the value of the cell is greater than 100.

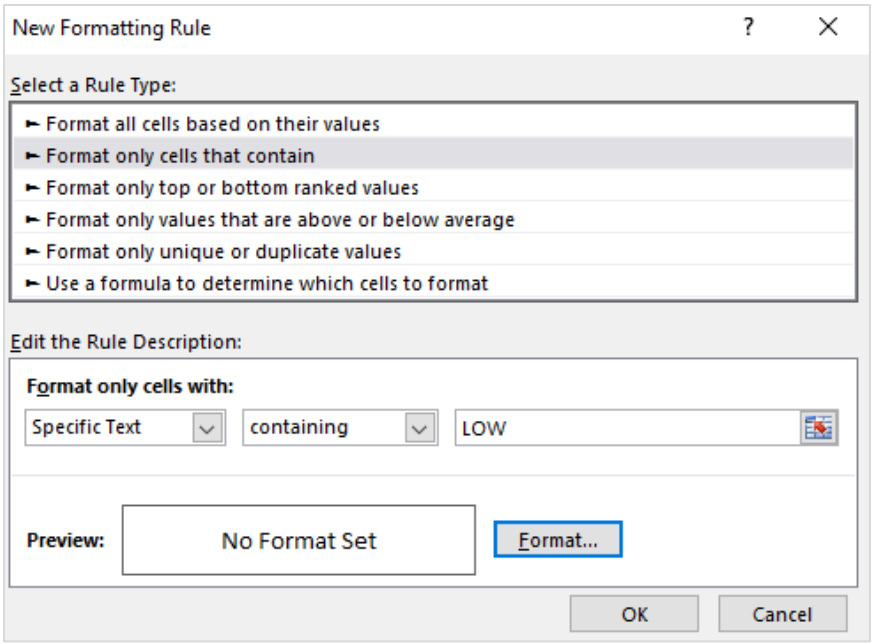


For our activity, we can change the font color to red if the display text is LOW and green if the display text is GOOD.

117. Select cells from **D5** to **D14.** With the cells selected, in the **Home** tab, click on the **Conditional Formatting** icon from the **Styles** group. From the **Conditional Formatting** list, move the mouse to the **New Rule…** and click on **New Rule…** icon.

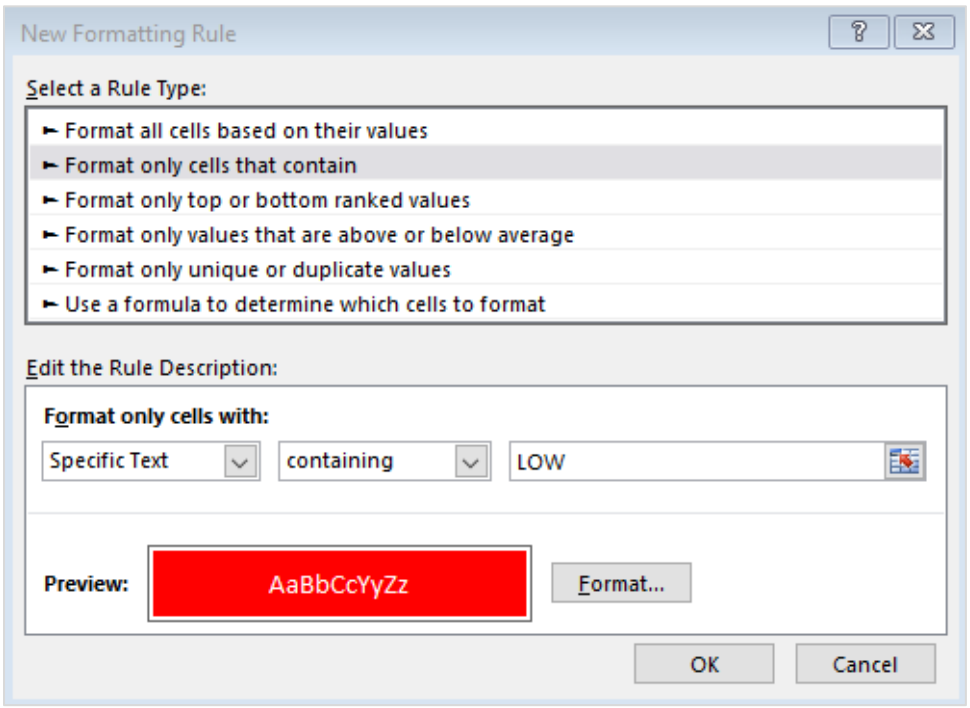


118. In the **New Formatting Rule** window, click on **Format only cells that contain** from the **Select a Rule Type** list.
119. From the **Edit the Rule Description** session, click on the **Cell Value** arrow and click on **Specific Text** choice.
120. On the right text field, type **LOW**



121. From the bottom of **New Formatting Rule** window, click on **Format…** button to open the **Format Cells** window.
122. In the **Format Cells** window, click on **Fill** tab, and select the **Red** color. Click on the **Font** tab, click on the **Color:** arrow and select the **White** color. Once we have the Fill and Font color selected, click on the **OK** button.

The **New Formatting Rule** should show a Preview of the cell:



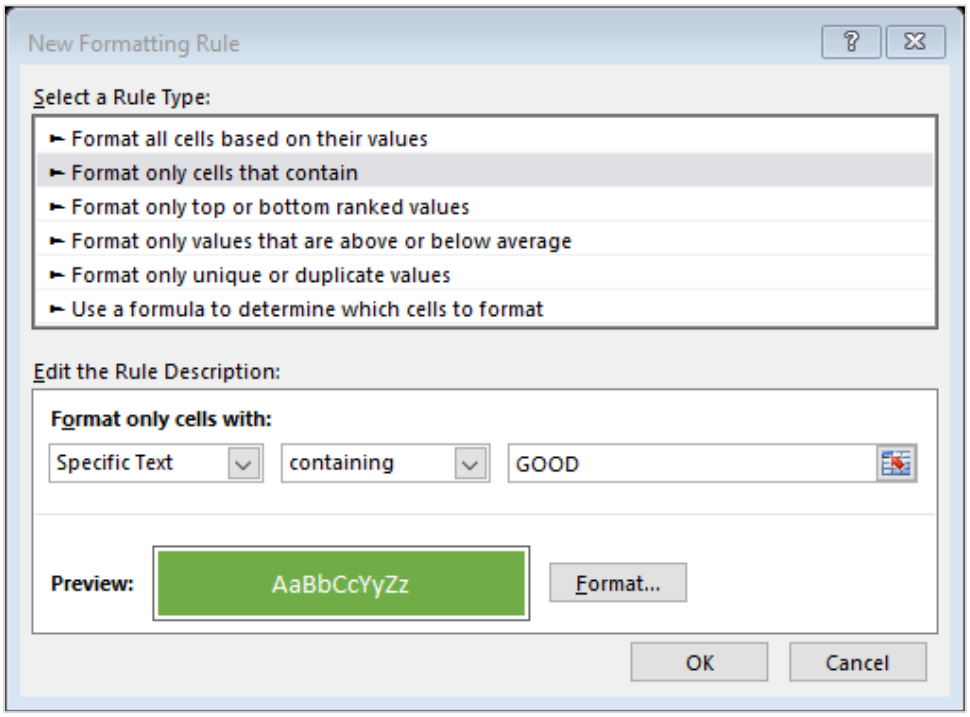
Once the new rule is already set, click on the **OK** button to close the **New Formatting Rule** window. The cells should be updated as the following image:



We can repeat the previous steps to change the cell color to green if the display is GOOD.

123. Select cells from **D5** to **D14.** With the cells selected, in the **Home** tab, click on the **Conditional Formatting** icon from the **Styles** group. From the **Conditional Formatting** list, move the mouse to the **New Rule…** and click on **New Rule…** icon.
124. In the **New Formatting Rule** window, click on **Format only cells that contain** from the **Select a Rule Type** list.
125. From the **Edit the Rule Description** session, click on the **Cell Value** arrow and click on **Specific Text** choice.
126. On the right text field, type **GOOD**
127. From the bottom of **New Formatting Rule** window, click on **Format…** button to open the **Format Cells** window.
128. In the **Format Cells** window, click on **Fill** tab, and select the **green** color. Click on the **Font** tab, click on the **Color:** arrow and select the **White** color. Once we have the Fill and Font color selected, click on the **OK** button.

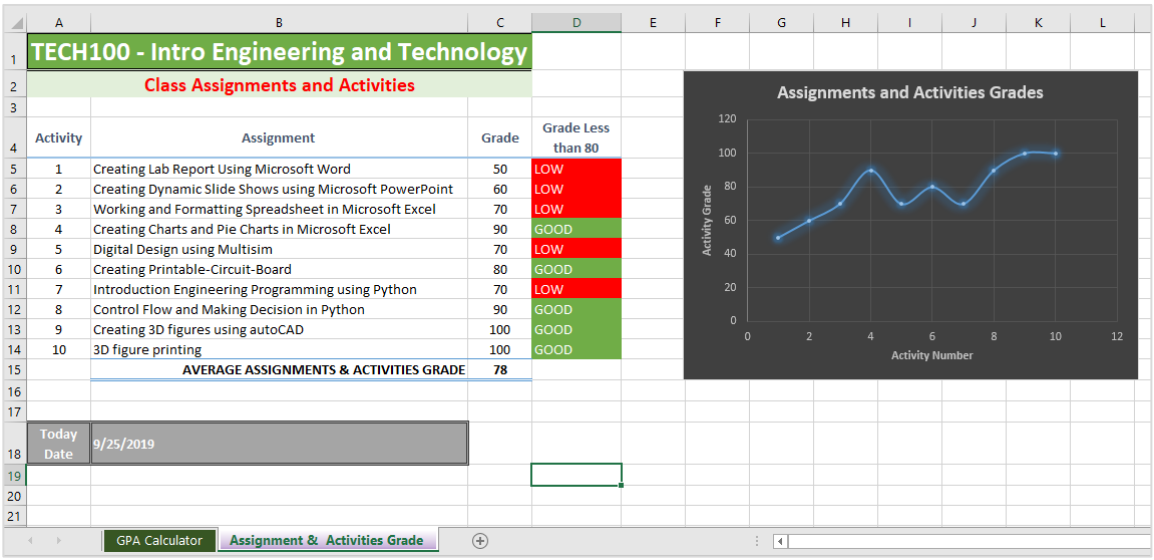
The **New Formatting Rule** should show a Preview of the cell:



Once the new rule is already set, click on the **OK** button to close the **New Formatting Rule** window. The cells should be updated as the following image:

Grade Less than 80
LOW
LOW
LOW
GOOD
LOW
GOOD
LOW
GOOD
GOOD
GOOD

Assignment & Activities Grade sheet should look as the following image:



	A	B	C	D
1	TECH100 - Intro Engineering and Technology			
2	Class Assignments and Activities			
3				
4	Activity	Assignment	Grade	Grade Less than 80
5	1	Creating Lab Report Using Microsoft Word	50	LOW
6	2	Creating Dynamic Slide Shows using Microsoft PowerPoint	60	LOW
7	3	Working and Formatting Spreadsheet in Microsoft Excel	70	LOW
8	4	Creating Charts and Pie Charts in Microsoft Excel	90	GOOD
9	5	Digital Design using Multisim	70	LOW
10	6	Creating Printable-Circuit-Board	80	GOOD
11	7	Introduction Engineering Programming using Python	70	LOW
12	8	Control Flow and Making Decision in Python	90	GOOD
13	9	Creating 3D figures using autoCAD	100	GOOD
14	10	3D figure printing	100	GOOD
15		AVERAGE ASSIGNMENTS & ACTIVITIES GRADE	78	
16				
17				
18	Today Date	9/25/2019		
19				
20				
21				

129. Save your work.

---------- Lab Experiment Ends Here ----------

Grading Policies

The lab is worth **20 points total**, distributed as follows:

- **Skill 1:** Creating and Formatting Cells and Spreadsheets (4 points)
- **Skill 2:** Using Mathematical Operators and Statistical Functions (4 points)
- **Skill 3:** Creating Charts and Pie Charts (4 points)
- **Skill 4:** Applying Functions in Excel (4 points)
- **Reflection Tab:** 4 points

To receive full credit, you must demonstrate all four skills (Skills 1–4) in your Excel spreadsheet. Each skill is worth 4 points.

In addition, you must create a **Reflection tab** in your spreadsheet that addresses the following questions:

A. What did you learn from this lab?

B. Which skill did you find the most difficult, and what steps did you take to complete it?

C. Did you use any Generative AI (GenAI) tools to complete any of the skills? If so, explain how you used them.

Formatting Requirement: The Reflection tab must include appropriate cell formatting.

References: Be sure to cite all materials and resources used to complete this lab. Failure to include proper references may result in a deduction of up to **5 points**.

Deliverables

Submit the completed Excel workbook in **.xlsx format** using the following file-naming convention:

Lab8StudentName.xlsx

Replace *StudentName* with your name before submitting the file.